Reproducing PatchTST

Vivien Chen, Katherine Hu, Michelle Zhou

1. Introduction

Time series forecasting is critical across energy, weather, and financial domains. RNNs and LSTMs struggle with parallelization over long sequences, while standard Transformers face quadratic attention complexity per timestep. We reproduce *A Time Series is Worth 64 Words: Long-term Forecasting with Transformers* (Nie et al., ICLR 2023), which introduces PatchTST. The model segments each channel's time series into patches and processes channels independently with shared weights. These design choices enable PatchTST to substantially outperform prior SOTA transformer-based forecasting models.

2. Chosen Result

We focused on reproducing the supervised multivariate long-term forecasting results reported in Table 3 of the original paper, of PatchTST/42 on various datasets forecasting electricity, illness, and weather. This table is the paper's central claim, directly demonstrating that patching and channel-independence outperform prior Transformer baselines. These datasets are standard benchmarks used across the forecasting literature, enabling direct comparison and validating generalizability across diverse prediction tasks.

Dataset	Paper MSE	Paper MAE
ETTh1	0.417	0.430
ETTh2	0.331	0.379
ETTm1	0.352	0.382
ETTm2	0.258	0.315
weather	0.229	0.265
ili	1.538	0.841

Figure 1. Table 3 results of the original paper

3. Methodology

Model Architecture

For each input sequence, we used a lookback window of 336 timesteps and processed each channel independently. The input sequence for each channel was segmented into overlapping patches(P=16, S=8) treated as singular tokens. Each patch is linearly projected into a latent embedding space of dimension d=128 with an added positional embedding. We applied Reversible Instance Normalization (RevIN) to each sample and channel.

The transformer encoder has 3 encoder layers (d=128), 16 attention heads, FFNNs (d=256), residual connections, BatchNorm, and dropout p=0.2. The model shares transformer weights across all channels. The output patch embeddings are flattened and passed through a linear prediction head that maps the latent representations to the forecasting horizon T.

Training and Evaluation
We evaluate on the datasets ETTh1, ETTh2, ETTm1, ETTm2, illness, and weather, with prediction horizons 96, 192, 336, 720. We had a train/test/validation split of 12/4/4 months. For our results, we used Mean Squared Error (MSE) and Mean Absolute Error (MAE). Training uses Adam (lr=1e-4, batch size 128). We train for 100 epochs with early stopping based on validation MSE.

Modifications
We tried two extensions of the original paper. First, we implemented cross-channel attention: patches from all channels were concatenated into a single token sequence with positional and channel embeddings, enabling attention across variables before reshaping back to per-channel outputs. We also implemented multi-scale patching, which processes the input in parallel at multiple patch sizes (8, 16, 32) and concatenates all results to create the prediction.

4. Results

Our reproduced results illustrated in Figure 2 are very close to the original paper across all datasets. Qualitative patterns are consistent with the paper, such as ETTh being harder to forecast than ETTm. Our MSE and MAE values are within 1-3% of the paper's figures, which can be attributed to variance in different random seeds and possible differences in small hyperparameters.

Dataset	Ours MSE	Paper MSE	Ours MAE	Paper MAE
ETTh1	0.419	0.417	0.431	0.430
ETTh2	0.346	0.331	0.391	0.379
ETTm1	0.348	0.352	0.378	0.382
ETTm2	0.256	0.258	0.316	0.315
weather	0.226	0.229	0.264	0.265
ili	1.974	1.538	0.958	0.841

Figure 2. MSE and MAE comparison of our reimplementation vs original paper

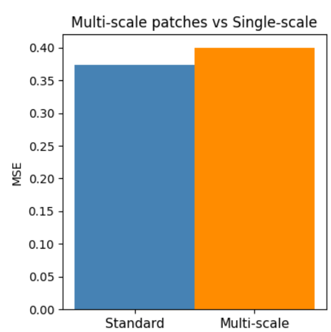

Figure 3. MSE comparison of multi-scale patching implementation vs standard

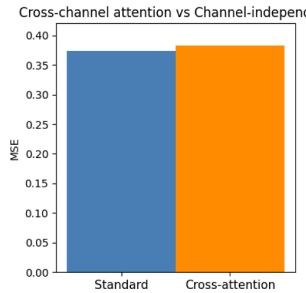

Figure 4. MSE comparison of cross-channel attention implementation vs standard

Figure 3 and 4 show the results of our extensions. Multi-scale patching allowed the model to capture finer temporal inflections but performed slightly worse overall. Cross-channel attention also underperformed the baseline, indicating that joint cross-variate attention introduces more spurious correlations than meaningful signal, reinforcing the paper's core claim that channel-independence is a strong inductive bias for this task.

5. Reflections

One challenge during reproduction was identifying important implementation details such as RevIN and dropout, which significantly affected performance. Initially omitting these components when we were trying to focus on the big picture led to noticeably worse results.
A key takeaway was that channel independence is more effective than expected. Although cross-channel attention seemed intuitively beneficial, it often introduced noise rather than useful information. We also found that patching improves more than computational efficiency. It enabled the model to learn broader temporal patterns instead of isolated timesteps.
Future work could explore better combinations of patch sizes to capture both small and large trends or selective cross-channel attention mechanisms applied only in specific transformer layers to introduce a good amount of meaningful information.

6. References

Nie, Y., Nguyen, N. H., Sinthong, P., & Kalagnanam, J. (2023). *A time series is worth 64 words: Long-term forecasting with transformers*. International Conference on Learning Representations (ICLR 2023).
We used Claude to create the skeleton template for our codebase and repository.